

Roshan Nepal

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<https://ee-roshan.github.io/RoshanNepal/>

Career Objective

A motivated and detail-oriented Electrical and Electronics Engineer with a strong foundation in control systems, robotics, and embedded systems. Passionate about applying innovative solutions in automation, power electronics, and machine learning. Seeking an internship/job to leverage technical expertise and managerial experience in a dynamic work environment.

Education

Bachelor of Engineering in Electrical and Electronics Engineering (Power and Control)

Kathmandu University, Dhulikhel

Expected Graduation: 2025

Relevant Courses: Control Systems, Power Electronics, Machine Learning, Embedded Systems, Robotics, Power Systems

Technical Skills

- Software & Tools: AutoCAD Electrical, DIALux, PowerFactory, ETAP
- Embedded Systems: Raspberry Pi, FPGA, Arduino, ESP32
- Simulation Tools: MATLAB/Simulink, Proteus, KiCad, LTspice
- Machine Learning: TensorFlow, Scikit-Learn
- Hardware: Sensors, LiDAR, Power Electronics
- Programming: Python, C, MATLAB, Simulink, C++

Projects

Line Following Robot

Design a line following robot using infrared sensors to detect and track a path, moving autonomously without a microcontroller.

Automatic Transfer Switch with Load Prioritization

Designed an IoT based automatic transfer switch(ATS) with load prioritization which seamlessly shifts power between two sources while intelligently managing the critical loads.

Static Var Compensator Prototype

Developed a compensation system to improve power factor in a single-phase system. Implemented a Fixed Capacitor – Thyristor Controlled Reactor.

Leadership & Managerial Experience

- Class Representative (CR) (4 Years) – Led and coordinated communication between students and faculty, managing academic and administrative concerns.
- Coordinator, EEPEX (Annual Technical Event) – Organized and managed technical events, guest lectures, and project exhibitions.

Languages

- English (Fluent)
- Nepali (Proficient)